

cold points. The hot points are rectified first, followed by the cold points.

Valor does calculations, too. DEK, a manufacturer of screen printing machines for SMT PCBs, needs to know precisely how much solder paste should be applied on the stencils; the transmission rate should be 70 per cent or more. Valor takes 24 to 48 hours to calculate, depending on the size of the project. Similarly, there are calculations required for some other processes, too.

Valor always has scripts running, and if a user wishes to add any process to it, a programmer will be required to modify the script. As per Saxena, Valor for electronics is a set of sophisticated DFM tools, but it does not have revised versions yet. However, if on-the-air changes are made in the software, it gets updated elsewhere simultaneously.

Valor users need to convert the SAP or ERP, CAD, PCAD files, or other PCB design and simulation files, into ODB ++ format, which is a readable format for Valor.

Some other tools

A tool called Apriori provides a distinctive platform for manufacturers to manage their product's cost at all stages of its development and manufacture. This starts right from design stage to manufacturing stage. Apriori focuses on how the project cost can be made feasible and enables quick iterations as per changing market, material cost and availability.

The tool called Project Tracker is a software that updates tracking of tasks, assignments, events and activities related to a project. This helps project managers in consistent project execution, improved resource utilisation, reduced resource gaps and operational efficiency. It facilitates communication and decision-making, so you can manage the project efficiently and deliver on time—making the whole process hassle-free and budget-friendly.

While Cisco considers Valor as mandatory, some other electronics

manufacturers carry out DFM using such PCB designing and simulation software as PCAD, CAD, PRO-E for electronics, or Solidworks for mechanical assemblies for sheet-metal checking, injection moulding, plastic mould filling, checking drafts, undercuts and thickness, etc.

New DFM tools such as Should Cost Analysis, Cost-Out Analysis and Fast Works for sourcing and purchase are being used by some established American electronics conglomerates.

Monitoring cost during design stage

According to an industry expert, Fast Works is useful for the development of new products in a company. It helps in getting close to the customer, increases chances of success, speeds up marketing and makes it easy to get things done. Usually, companies use Fast Work for costing during new product development followed by the launch of prototype in the market.

There are stages of this methodology which define team work, focusing on the customer's weak points, finding solution to it and documenting the whole process. After all, early learning and validation hypothesis is the key to success. Besides, the team needs to reduce the 'build-measure-learn' time loop. Above all, experiments and quick iterations as per customers' requirements also need to be done.

Cost savors during running of product

Should Cost Analysis sets an accurate benchmark for cost negotiation. Designers get the clarity of vendor pricing through this tool, thus helping them make an informed decision regarding the cost of components. This helps the designer to cut down the cost during the design stage itself. Apriori and SAP are the two tools that can be used for Should Cost Analysis.

Should Cost Analysis covers technology and quality, customer satisfaction, price and serviceability. Companies use either Should Cost

Analysis or Cost-Out Analysis for a product. After the analysis, the designer may suggest the customer to forget the existing solution and undertake a new project, or may propose an additional feature for the product. This is how design for manufacturing is managed from the initial to its final stage.

DFM sans tools

While you may think that these tools are essential for the success of DFM, fortunately that is not the case. According to industry experts, tools like Valor, Apriori, SAP and Project Tracker are very useful, but some of these are actually very demanding, and require a huge one-time investment and rigorous training. Not all small-scale manufacturing units or startups can afford to purchase these. But a third-party testing is always an option.

As an alternative, a rulebook for DFM can be made by a panel of experienced designers and manufacturers, which can be used by the design engineers. Apart from using Valor, Cisco has a handy rulebook that it follows. Similarly, all other electronics manufacturing MNCs also, besides using DFM tools, follow their handy rulebooks that are prepared in-house.

The Indian story

We have all heard horror stories related to Indian manufacturing sector. Most companies have been using certain DFM tools without complete knowledge about them. In some other companies design engineers are unaware that they have been using DFM tools, thus being unable to realise their full potential. This is happening because of lack of requisite training.

Many small and medium enterprises do not use DFM because they do not realise its importance. Thus, they lose crucial time for design and manufacture. A company should always be ready to keep its design team on toes with new techniques, trainings and software. Perfect design and proper costing is like the heart and soul of a perfect product. **EFY**